

The Living Map: Spatial Memory for Emergency Robots

Short technical report – IEEE TSYP14, Phase 1 – Fire / Hazardous Building

ABSTRACT

We present a system in which a *Writer* robot explores a GPS-denied burning building, decides what is worth recording and leaves the answer in the building as 32-byte radio *beacons*. A van (the Outside Network Area) is the only link between inside and outside: it decodes the beacons, rebuilds their positions from egocentric hops, converts them to GPS and forwards them to a Command Post that ranks, groups and assigns the events to *Executor* robots. Robots never talk to each other or to the Command Post. On a simulated 48×24 m floor (10 seeds) the system detects 100 % of the events with 0 false records, locates them to 0.42 m on average, and delivers every beacon in under one second. A briefed Executor reaches the first person in 76 s instead of 172 s for a blind one, but the end-to-end chain is slower (303 s) because the Writer must explore first. We report five failure cases, including two where the design *fails* (Writer lost; strong wall attenuation). All numbers come from simulation with assumed radio and sensor parameters.

1. PROBLEM AND APPROACH

Inside a burning building there is no GPS, little radio, and no time. A robot sent in blind must search, and every second in the heat costs sensors and lives. Our idea is to separate *finding* from *acting*. One Writer robot goes in first and writes what it finds into the building itself; everything downstream (ranking, dispatch, navigation) works from that memory. Design rules: (i) the Writer records only corroborated events, so the memory is small; (ii) positions are stored *relative* to each other and translated to GPS only outside; (iii) a star topology: robots only ever talk to the van; (iv) no single beacon is a single point of failure.

2. SYSTEM ARCHITECTURE

Fig. 1 shows the three zones: inside (ROS domain 10), the van, and the Command Post (domain 20). The two domains share no node or topic; the only crossings are the UDP uplink (port 9101), the UDP downlink (9102) and the briefing file handed over at the van before an Executor enters. The code has four layers: plain-Python core logic (`beacon_protocol`, `geo`, `sim_core`, `adaptive_queue`, 2547 lines in total for the package), actors (`agents`, `command_post`, `roles`), three orchestrators (one-process simulator with web pages, ROS 2 nodes, Gazebo world generator) and tooling. *The core never imports ROS*; the ROS nodes are thin wrappers, so the same code runs in the simulator, in unit tests (47 tests) and on ROS 2. `audit.py` and a test fail if any robot subscribes to anything other than its own van topic.

3. WRITER AUTONOMY, DETECTION AND DEPOSITION

Approach and exploration. The Writer starts at the van, outside, where GNSS works. It drives to the entrance door steering by noisy fixes ($\sigma=0.8$ m, assumed); in the last 2.5 m the LiDAR finds the door gap. At the threshold GPS is lost and the private frame is anchored on the last fix; from there only LiDAR and odometry (drift 1 %) are used. The Writer keeps its own occupancy map and repeatedly plans an A* path to the nearest frontier cell within 60 m; cells near heat are penalised (+40 path cost). When no frontier is left it reports `EXPLORATION_DONE` through the van and drives back to park at the van. This is our own exploration and localisation model, not `slam_toolbox/Nav2`; porting to them is a Phase 2 item.

Event detection. A 2-of-3 rule requires two sensor votes among air temperature ($>70^\circ\text{C}$), thermal-camera maximum ($>150^\circ\text{C}$) and gas (>50) for 3 consecutive readings before a fire is declared. A steam pipe raises air temperature but not the thermal signature; one vote only, so it is counted as `steam_rejected` and never beacons. Gas is *tracked*: after the threshold (150 ppm) is crossed the Writer

follows the plume for up to 25 s and writes the beacon at the *peak*, not at first contact. People and animals come from thermal blobs (confidence ≥ 0.6 , persistence 3). A person within 3 m of a hot spot, or in gas above 500 ppm, gets class P0. If LiDAR sees rubble on the line of sight to a person, the Writer drops a `DEBRIS` beacon and flags the person `human_intervention_required`. Thresholds are per-map parameters.

Beacon deposition. Beacons are limited to a budget of 44. The Writer chooses the parent with fewest hops among nodes whose link is at least -72 dBm and within 12 m (the hop field is limited to ± 12.7 m). If only links between -80 and -72 dBm remain it drops a `REPEATER` *before* the link breaks, keeping the mesh dense. Landmarks (stairs, shafts) are beacons so that other robots can change floor.

4. BEACON MESSAGE AND SIGNAL DESIGN

Frame. The beacon v2 is exactly 32 bytes (checked by an assertion at import, Fig. 2): 27 bytes of body (magic `0xB6`, id, origin, version, t_0 , `typeidclassflags`, floor, victim count n , heading, hop and target offsets, parent id, temperature, gas, mirror id and sequence), a CRC16-CCITT (2 B) and an HMAC-SHA256 truncated to 3 B. The packet type has 4 bits (repeater, fire, gas, human, animal, debris, beacon-lost, stairs), the priority class 2 bits (P0 critical to P3 routine), the flags 2 bits (`human_intervention_required`, verified).

Egocentric geometry. A beacon never stores absolute coordinates. It says “I am h_f m ahead and h_r m to the right of my parent, in the parent’s heading; the target is t_f, t_r from me.” Heading $\theta = 2\pi q/256$; forward = $(\cos \theta, \sin \theta)$, right = $(\sin \theta, -\cos \theta)$. The van chains hops from the entrance, so each hop is measured locally and the error grows with the *number of hops*, not the path travelled. Decimetre resolution over ± 12.7 m fits one signed byte.

Integrity and trust. CRC16 detects corruption; the 3-byte MAC rejects forged packets (24 bits: a deterrent, not strong authentication; the key is a demo constant, to be per-deployment on hardware). Duplicate (*id*,*seq*) pairs are dropped.

Resilience. Each beacon carries a neighbour’s id and sequence (*mirror frame*), and neighbours keep copies (gossip, RSSI ≥ -90 dBm). A beacon silent for 15 s is re-broadcast by the best neighbour as `BEACON_LOST` (id $1000+id$, class P3): silence is data. Packets that did not reach the van are retried every 5 s (ARQ, 3 tries per hop) and ancestors lost earlier are forwarded first (store-and-forward). A robot can also write a mutable state byte on a beacon over short range; the signed report is never changed.

The Living Map: system architecture (star topology, one crossing)

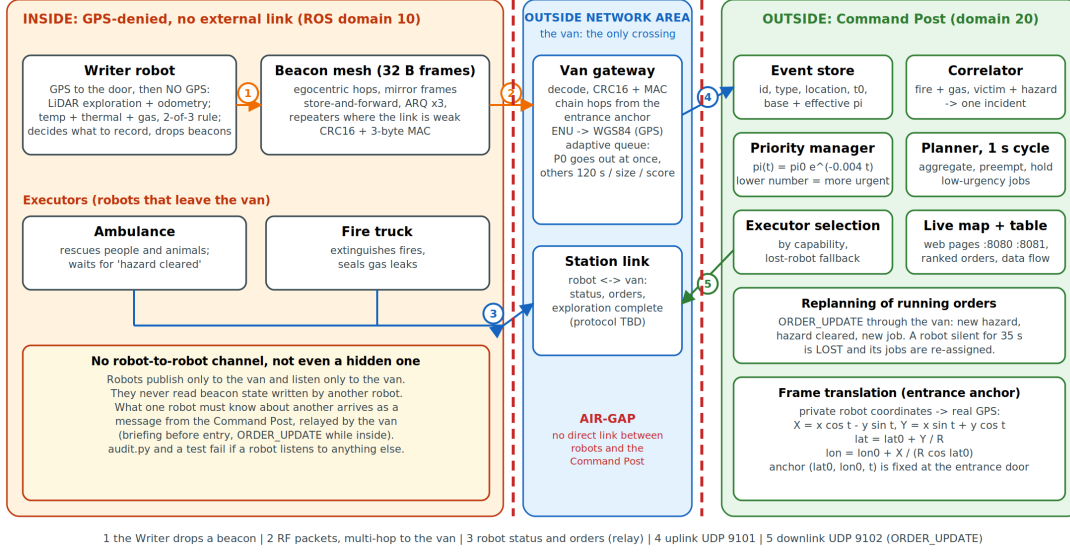


Figure 1: Architecture: (1) the Writer drops a beacon; (2) multi-hop RF to the van; (3) robot status and orders relayed by the van; (4) UDP uplink; (5) UDP downlink (ORDER_UPDATE).

Beacon v2: 32 bytes, egocentric (no absolute coordinates inside the building)

bytes 0-15										
magic 1 B	id 2 B beacon	org 1 B	seq 2 B version	t0 4 B written at (s)	T P F 1 B	floor 1 B	n 1 B	hdg 1 B	hopF 1 B	hopR 1 B
bytes 16-31										
tgtF 1 B	tgtR 1 B	parent 2 B chain	temp 2 B 0.1 C	gas 2 B ppm	mirror id 2 B neighbour	m.seq 1 B	CRC16 2 B integrity	MAC 3 B HMAC-SHA256/24		

T|P|F = packet type (4 bits: REPEATER, FIRE, GAS, HUMAN, ANIMAL, DEBRIS, BEACON_LOST, STAIRS), priority class P0..P3 (2 bits), flags (2 bits: human_intervention_required, verified). hdg = heading in 1/256 turns. hopF/hopR/tgtF/tgtR = metres ahead / to the right in decimetres (+12.7 m). temp = 0.1 C units. gas = ppm, n = number of victims, t0 = time the beacon was written (used for aging), mirror = id + seq of a neighbour.

How positions are rebuilt: each beacon says 'I am hopF m ahead and hopR m to the right of my PARENT (in the parent's heading); what I found is tgtF / tgtR from ME'. The van chains the hops from the entrance anchor, so Writer drift does not accumulate in the stored frames. Mirror frames: each beacon also carries a neighbour's id + seq, so a destroyed beacon is detected (missing heartbeat) and its data recovered.

Figure 2: Beacon v2 layout (32 bytes) and how positions are rebuilt by chaining hops.

Radio model. $RSSI = -40 - 28 \log_{10} d - L_w n_w - 4 n_r - 18 n_f$ dB (n_w walls, n_r rubble piles, n_f floors; $L_w = 12$ dB by default). Packet-error rate is 0 above -85 dBm, rises linearly to 0.9 at -95 dBm, and is 1 below. A 32-byte frame at SF7 (433/868 MHz LoRa, 125 kHz) is 72 ms on air, which allows 500 frames per hour at a 1 % duty cycle (our computation).

5. FRAME TRANSLATION

The anchor $(\phi_0, \lambda_0, \theta_0)$ is the entrance door. A local point (x, y) is rotated to east/north,

$$E = x \cos \theta_0 - y \sin \theta_0, \quad N = x \sin \theta_0 + y \cos \theta_0,$$

and converted with the WGS84 radii of curvature M, N_ϕ at ϕ_0 :

$$\phi = \phi_0 + \frac{180}{\pi} \frac{N}{M}, \quad \lambda = \lambda_0 + \frac{180}{\pi} \frac{E}{N_\phi \cos \phi_0}.$$

The inverse is used when an Executor reads a briefing. Each record carries an uncertainty $\sigma = \sqrt{\sigma_a^2 + (kd)^2 + (d\sigma_\theta)^2}$ with $\sigma_a = 0.5$ m (anchor), $k = 0.02$ (SLAM scale), $\sigma_\theta = 2^\circ$, d the distance from the entrance (e.g. 1.7 m at 40 m; assumed values). Measured error in simulation is in Section 9.

6. OUTSIDE NETWORK AREA (THE VAN)

The van antenna receives packets, verifies CRC/MAC, orders them by parent (a child waiting for an unseen parent is held), chains the hops, converts to GPS and enqueues the

record. A fire or gas beacon within 6 m of a P0 person escalates to P0. The *adaptive queue* sends P0 at once together with its ancestor chain and nearby hazards (so the Command Post can read it); other records flush after 120 s, at 8 items, or when the queue score $Q = U_0 C (1 + 0.01 w)$ reaches 20 (C : confidence, w : waiting time). Default mode is *immediate* (every record forwarded); *batched* is the bandwidth-saving mode. The van also relays robot status (STATUS, ARRIVED, RESCUED, FIRE_OUT, UNREACHABLE, ...) and Command Post ORDER_UPDATES. The robot-van radio protocol is an ideal channel in simulation (framing/ARQ to be defined in Phase 2).

7. COMMAND POST

Two clocks. *Confidence* falls with age, $C = e^{-\lambda \text{age}/60}$ ($\lambda = 0.10/\text{min}$ fire and gas, 0.01 people and animals); records with $C < 0.5$ and class $\leq P1$ are flagged for re-verification. The *priority number* $\pi \in [0, \infty)$, *lower = more urgent*, falls while an event waits,

$$\pi(t) = \pi_0 e^{-0.004t}, \quad \pi_0 = \frac{\text{base}(\text{type}) \cdot f(\text{class})}{\max(n, 1)},$$

base: person 3, fire 5, gas 6, animal 10, debris 30, lost beacon 60; f : P0 0.15, P1 1, P2 1.5, P3 2. Nothing waits forever (half as heavy after ≈ 3 min), and age counts from detection,

End-to-end sequence: from detection to a finished rescue

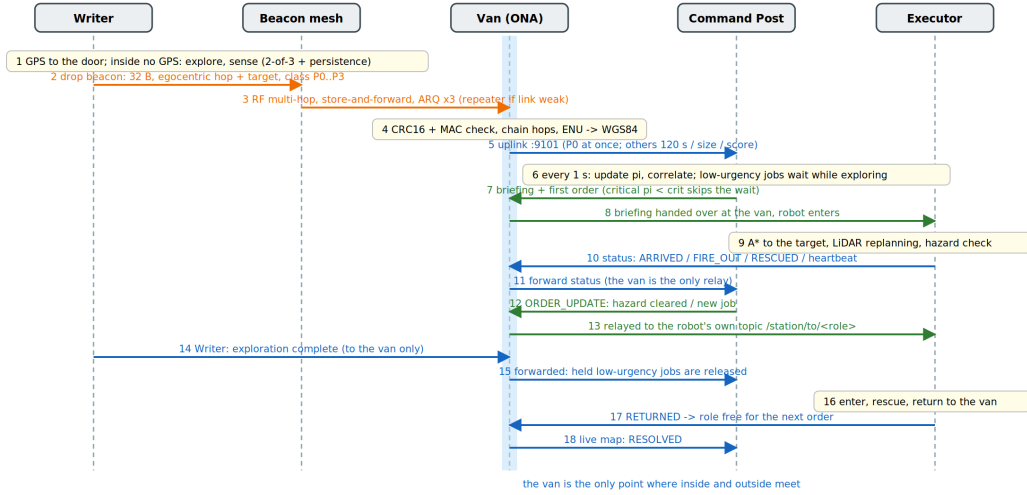


Figure 3: Sequence of one mission: GPS approach, detection, beacon, uplink, order, briefing at the van, Executor action, ORDER_UPDATE and status messages.

not last report. A hazard in the same incident as a victim inherits 0.9π of that victim.

Decision cycle. Once per second (not once per event) the Command Post updates priorities, correlates events into *incidents* (union-find: at least one hazard, same floor, ≤ 8 m, ≤ 300 s apart; transitive, so fire-gas-person is one incident), aggregates, preempts and issues *one order per robot*. A critical event ($\pi < 0.3$) skips the collection wait. Executors are chosen by *capability* (extinguish, seal gas, rescue), so a lost robot (silent 35 s) can be replaced. Running orders are updated through the van.

Hold rule. A job whose π is worse than the least urgent life-threatening event at class P1 (computed: 6.0) does not send a robot out while the Writer is exploring; it is released by aging or at once on EXPLORATION_DONE. Without it the ambulance leaves for an animal at 87 s; with it, at 228 s, after a person is known (mission 384 vs 337 s, victims rescued 2.0 in both).

8. EXECUTOR ROBOTS

An Executor (ambulance: rescue; fire truck: extinguish and seal gas) receives a briefing at the van: targets with GPS and π , the beacon chain as a path hint, hazards with radius (fire 2.5 m, gas 3.0 m), debris and skipped targets. It plans A* straight to the target with a free-space assumption (unknown space passable) and uses the Writer's trail only as a safe hint. Multiple jobs are ordered to minimise $\sum_i T_i / \pi_i$ (travel at robot speed, 15 s service, 25 s per floor change; exact for ≤ 7 jobs). A victim in a danger zone is entered only after the Command Post reports the hazard cleared (150 s timeout). Targets flagged *human_intervention_required* are skipped. If no route exists the Executor reports UNREACHABLE; a target failed twice is dropped. Executors never read state written by other robots; what one must know of another arrives as a Command Post message.

9. RESULTS (SIMULATION)

Setup. Map: one floor, 48×24 m, 0.5 m cells, 2 fires, a gas leak, a steam source, 3 people (one sealed behind rubble), an animal. 10 seeds; faults 3 seeds; radio levels 2 seeds. Reproduce with `python3 tools/collect_final.py`.

Table 1 and Fig. 6a give the main numbers. *Read the tim-*

Table 1: Baseline on the building (10 seeds, mean $\pm$ s.d. where given).

Detection	recall 100 %; 0 false records in 10 runs
Position error	0.42 m mean, 1.38 m worst (person 0.30, animal 0.09, fire 0.13, gas 1.28 m, seed 1)
Beacons	20.8 per run (7 events + 13.8 repeaters), 100 % delivered; latency 0.06 s mean, 0.5 s max
Orders	4 orders and 6 updates for 7 event beacons
Outcome	2 of 2 fires out; 2 of 3 people rescued, the third (sealed) handed to humans in 10/10 runs
Timeline	GPS approach 3.5 s; first order 50 s; ambulance leaves 228 s; Writer done 284 s; fires out 287 s; mission 384.5 ± 6.9 s
First person	blind 172 s; briefed 76.3 ± 7.6 s; end-to-end 302.9 ± 6.4 s
Speed	$37\times$ real time

ing honestly: a briefed Executor needs 56 % less time than a blind one, but because the Writer must explore first, the first person is reached after 303 s, *slower than a lucky blind search*. The value of the chain is that no robot searches blind in the hazard, one Writer pass serves every Executor, the gas leak and fires are known before anyone enters, and a sealed victim is handed to the human team. Direct A* beats replaying the trail (first person 72 vs 122 s; distance 284 vs 319 m, 4 seeds). Batching cuts uplink messages from 20.7 to 6.0 per run at the cost of delay for P1–P3 (Fig. 6b); P0 is never held (0.25 s).

10. FAILURE CASES

Table 2: Failure cases: mitigation and measured outcome (building).

Case	Mitigation	Outcome
1 Beacon destroyed (120 s)	mirror frames; neighbour sends BEACON_LOST	reported 15.5 s later; mission unchanged (2/2 fires out)
2 Writer lost (150 s)	van forwards at once; Command Post keeps memory	0/2 fires out, 1.0/3 rescued ; recall 0.50
3 Steam, priority, batching	2-of-3 rule; P0 bypasses queue	0 false records; P0 delay 0.25 s
4 Impassable debris	debris beacon + human flag; Executors skip	flagged in 10/10 runs; no robot stuck
5 RF attenuation	repeaters, ARQ, store-and-forward	fine to 24 dB/wall; fails at 30 dB (1 of 2 seeds)

Writer lost. Everything reported before 150 s is already at the Command Post, so Executors still handle the gas

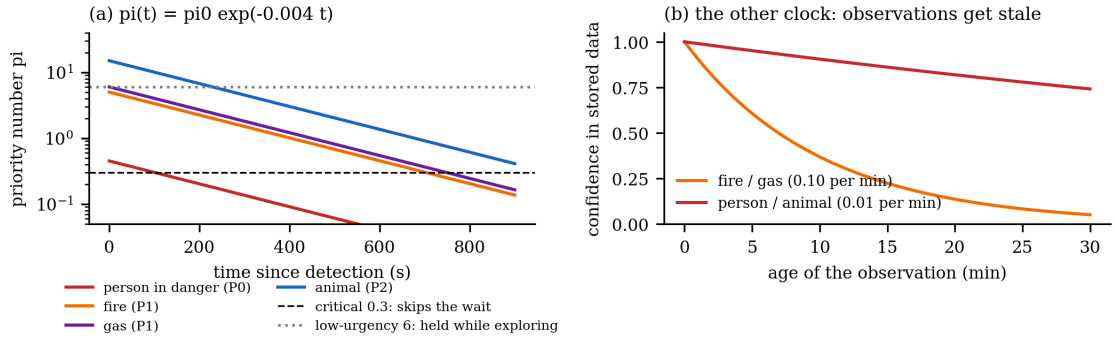


Figure 4: The two clocks: confidence falls with age while the priority number π falls (more urgent) as an event waits.

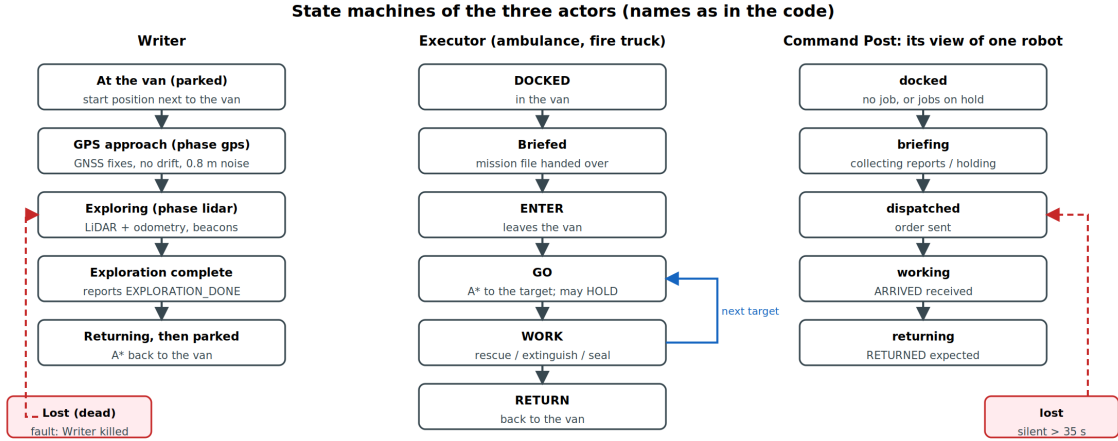


Figure 5: State machines of the Writer, the Executor and the Command Post decision cycle.

leak and what was found. But both fires are in the east part the Writer had not reached; none is extinguished and EXPLORATION_DONE never arrives (low-urgency jobs are released by aging). This is the honest cost of a single scout; a second Writer or an Executor that also maps would remove it. **Steam** is rejected by the detector (unit test; in the building the Writer never passes the source, so 0 rejections).

RF attenuation. Loss per wall L_w : 12 / 24 / 30 dB gives fires out 4/4, 4/4, 2/4; repeaters 13.5 / 21.5 / 21.0; delivered 20.5/20.5, 28.5/28.5, 14.5/28.5. At 30 dB the Writer crosses many walls and each repeater spends part of the 44-beacon budget, so too little reaches the Command Post. NIST 1997 data at 2.4 GHz give 29–31 dB for 203 mm (reinforced) concrete: *the design fails in heavy concrete*. Mitigations (not implemented): sub-GHz LoRa, more relay nodes, a budget reserved for events.

11. VERIFICATION AND REPRODUCIBILITY

The package has 47 pytest tests covering the protocol (pack/unpack round trip, CRC and MAC rejection), geometry and frame translation, the detector (including `test_detector_rejects_steam`), the mesh (store-and-forward, `BEACON_LOST`), the gateway, Command Post rules (aging, correlation, preemption, hold), navigation, the GPS approach, plus `test_audit_forbids_robots_reading_beacon_state`.

`test/fake_ros_integration.py` runs the ROS layer against a fake `rcipy` with real UDP and HTTP and checks the star topology and order updates. Every number in this report is produced by one script (`tools/collect_final.py`, seeded and resumable, output `docs/results_final.json`); the figures and documents are regenerated from that file. A measurement bug in v4_15 (a sortie ending in the same step as the next began was not recorded, over-reporting the Executor's first-person time) was found and fixed; earlier two-floor numbers are obsolete and not used here. Three web pages show the running system: the simulation (port 8081), the Command Post (8080) and the data flow with the beacon table (8081/flow).

12. LIMITS AND PHASE 2 PLAN

Limits. Everything is simulated; radio loss, GNSS error, sensor noise and thermal behaviour are *assumed*. The robot–van link is ideal. The Gazebo world is validated only as well-formed XML. The 24-bit MAC and demo key are not production security. Preemption is covered by unit tests, not by these scenarios.

Plan (6 Oct – 1 Dec). Hardware: Writer + one universal Executor (rescue, suppress, seal; the Command Post still assumes separate roles), 6–8 ESP32-S3 + LoRa beacon nodes running the 32-byte frame, LiDAR, thermal camera (Lepton), MQ-4 gas sensor (preheat ≥ 24 h), IMU, GNSS, Jetson Orin Nano; the DWM1000 UWB module is end-of-life, so UWB is optional (DWM3001CDK) and LoRa carries messages. Software gaps: universal Executor, robot–van protocol, beacon firmware with per-deployment keys, sensor drivers, RF characterisation, Gazebo validation. Ten physical tests (T1–T10) cover exploration, detection, beacon

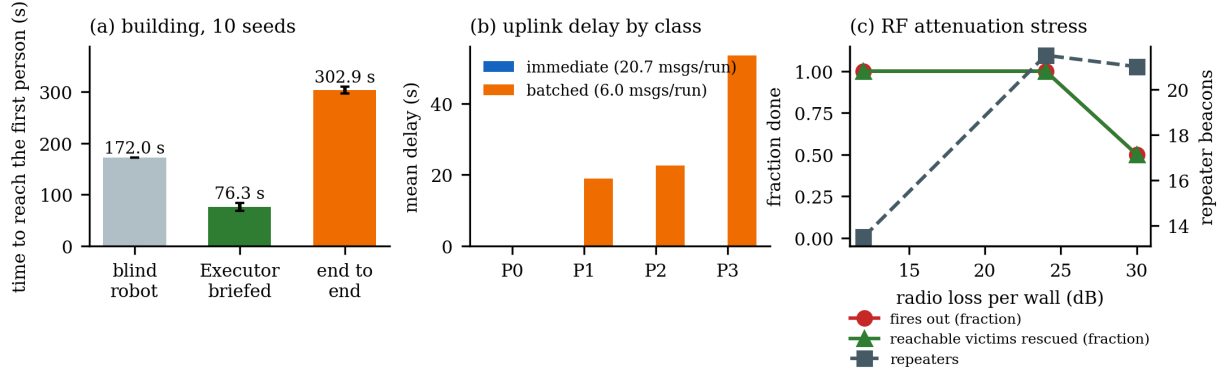


Figure 6: (a) time to the first person; (b) mean uplink delay by class (immediate delays are below 0.3 s); (c) RF stress: fraction of fires out/reachable victims rescued and repeaters used versus loss per wall.

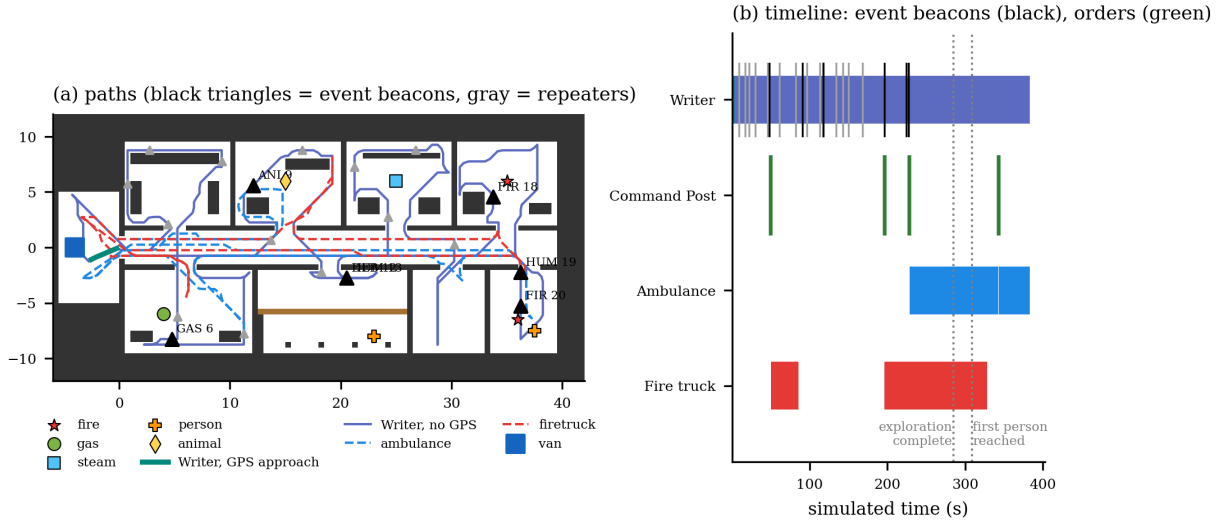


Figure 7: One traced run: (a) paths of the Writer (GPS approach in green), ambulance and fire truck, beacons and events; (b) timeline of event beacons (black), repeaters (grey), orders (green) and Executor sorties.

on air, air-gap, frame translation, briefed Executor, the five failure cases, RF shield, batching and three end-to-end runs.

REFERENCES

- [1] NIST (1997) indoor RF attenuation measurements at 2.4 GHz, as tabulated by Wi-Fi Vitae.
- [2] LoRa airtime for SF7–SF12 computed from the Semtech modulation